ASHESI UNIVERSITY

STYLEMATCH AI

An Explainable Position-Aware Football Player Profiling, Similarity and Recruitment Decision-Support System

FINAL PROJECT REPORT

CS 254 - Introduction to Artificial Intelligence

Group 14 | May-August 2026

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GitHub: github.com/wunnam-haruna/INTRO-TO-AI-GROUP-14

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Abstract

Football recruitment increasingly relies on data, yet player comparisons are often distorted by differences in position, minutes played, and tactical context. This project developed StyleMatch AI, an explainable position-aware football profiling and playstyle-similarity application. A 2025-2026 dataset covering 2,839 players from Europe's five major leagues was cleaned, converted to per-90 statistics, filtered at a 900-minute threshold, and standardised within broad position groups. Separate K-Means models were evaluated for defenders, midfielders, and forwards using silhouette, Calinski-Harabasz, Davies-Bouldin, inertia, cluster balance, and stability criteria. The final baseline selected two clusters per position and produced six interpretable archetypes. Across eight random seeds, all position models achieved an Adjusted Rand Index of 1.0, indicating reproducible assignments. A Euclidean-distance similarity engine and Streamlit game transformed the model into an interactive prototype in which users identify statistically similar players and receive model-based feedback. The system demonstrates unsupervised learning while foregrounding transparency, bias, and responsible human oversight.

1. Introduction

Football clubs, academies and managers make high-stakes decisions about player selection, replacement and development. Although performance data can support these decisions, simple rankings frequently ignore a basic football reality: the meaning of a statistic depends on the player's role. A forward is expected to shoot and score more often than a defender, while interceptions and tackles may be more informative for defensive or ball-winning roles. Comparing all players with one undifferentiated metric can therefore reward playing time, reputation or position-specific volume rather than genuine stylistic similarity.

StyleMatch AI addresses this problem by grouping players within broad position categories and discovering recurring statistical profiles through unsupervised learning. The system combines a reproducible data pipeline, position-based standardisation, K-Means clustering, multi-metric validation, a similarity engine and an interactive game. The game presents one target player and four candidates, then rewards the user according to how closely the chosen candidate matches the strongest model-identified alternative. This converts the model from a static analysis into a working human-AI decision-support prototype.

1.1 Objectives

- Prepare a reliable player-level dataset that controls for playing time and positional differences.
- Discover interpretable outfield player archetypes using separate K-Means models for defenders, midfielders and forwards.
- Evaluate cluster quality and reproducibility using several complementary validation measures.
- Build a similarity engine that retrieves statistically comparable players from the same broad position.
- Deploy the model through an interactive StyleMatch AI game that is understandable to technical and non-technical users.
- Audit the system for bias, fairness, privacy, transparency and responsible human oversight.

2. Background and Related Work

Clustering is well suited to football analytics because labelled tactical roles are often unavailable or subjective. K-Means partitions observations into groups by minimising within-cluster squared distance, making it useful for discovering latent patterns in standardised performance data (MacQueen, 1967). However, football roles cannot be inferred credibly from mathematical separation alone; clusters must also be stable, reasonably balanced and interpretable in domain terms.

Prior research supports the use of unsupervised methods for playing-style discovery. Diquigiovanni and Scarpa (2019) grouped association-football styles using clustered network structures, while Akhanli and Hennig (2023) clustered European player-performance data and emphasised that the number and nature of clusters should be chosen according to the intended use. Their study is especially relevant because it distinguishes broad player groupings from smaller similarity-oriented clusters. A recent systematic review similarly concludes that unsupervised learning can support tactical analysis and recruitment, but highlights inconsistent preprocessing, feature selection and reporting as major threats to reproducibility (Gu et al., 2025).

StyleMatch AI responds to these concerns by making preprocessing explicit, fitting models separately by position, testing candidate K values, reporting several validity indices and separating model stability from realworld validity. The project also adds an interactive layer: rather than presenting a black-box recommendation, it reveals candidate rankings, relative match quality and the statistical basis of each comparison.

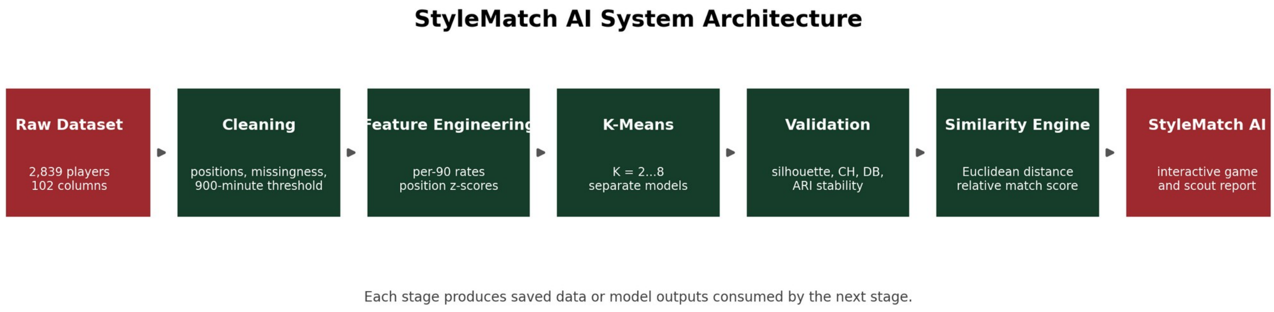

Figure 1. End-to-end architecture of the StyleMatch AI system.

3. Methodology

3.1 Data Source and Unit of Analysis

The project used the Football Players Stats 2025-2026 Kaggle dataset published by Hubert Sidorowicz and sourced from FBref. The downloaded full file contained 2,839 player records and 102 columns across the Premier League, La Liga, Serie A, Bundesliga and Ligue 1 (Sidorowicz, 2026). One observation represented a player-club-competition record for the season. The dataset contained no exact duplicate rows and no duplicate Player-Squad-Competition combinations.

Audit Measure	Verified Result
Raw records	2,839
Raw variables	102
Multi-position records	620
Players with at least 450 minutes	1,999
Players with at least 900 minutes	1,578
Players with at least 1,350 minutes	1,192
Eligible outfield players	1,464
Eligible goalkeepers	114

3.2 Position and Playing-Time Controls

The raw position field included single and dual labels such as DF, MF, FW, MF-FW and DF-MF. The first listed value was retained as the primary position; the second was stored separately, and a multi-position indicator was created. Primary labels were mapped to Goalkeeper, Defender, Midfielder and Forward. Goalkeepers were excluded from the outfield clustering because their performance variables are structurally different.

A 900-minute threshold was selected for the baseline model. This excludes highly volatile small samples while retaining 1,464 outfield players. Sensitivity counts were also documented at 450 and 1,350 minutes so that the threshold decision remained transparent.

3.3 Per-90 Transformation and Position Standardisation

Raw totals were converted to rates to reduce playing-time bias:

$$\text{Statistic per 90} = (\text{Total statistic} / \text{Minutes played}) \times 90$$

Twelve outfield features were retained: goals, assists, non-penalty goals, shots, shots on target, crosses, fouls drawn, tackles won, interceptions, fouls committed, offsides and yellow cards, all expressed per 90 minutes. The final modelling dataset had no missing values in these features. Where imputation was required in the reusable pipeline, medians were calculated separately within each position group.

z = (x - mean_position) / standard deviation_position

StandardScaler was fitted separately for defenders, midfielders and forwards. Consequently, a positive z-score means higher statistical output relative to the player's positional peers, not necessarily superior overall football quality.

3.4 Model Training and Selection

Separate K-Means models were tested for K = 2 through K = 8. Each candidate used 50 initialisations and random state 42. The final model used 100 initialisations. Selection considered silhouette score (cohesion and separation), Calinski-Harabasz score, Davies-Bouldin score, inertia, minimum cluster size and cluster-balance ratio. The silhouette measure follows Rousseeuw (1987), while repeated-run agreement was assessed with the Adjusted Rand Index of Hubert and Arabie (1985).

The model-selection rule prioritised silhouette performance, then Calinski-Harabasz and Davies-Bouldin rankings, while rejecting very small clusters where possible. Football interpretability was treated as an additional requirement: a mathematically generated cluster was accepted only when its centroid could be described meaningfully.

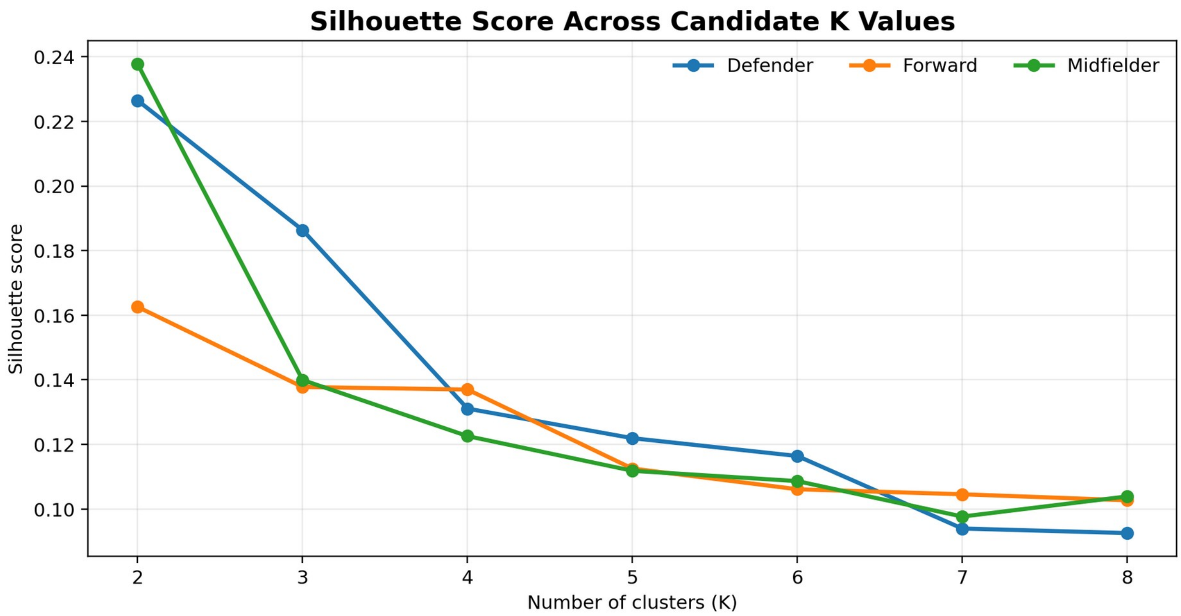

Figure 2. Silhouette scores for candidate K values. K = 2 was strongest for all three broad positions.

4. Implementation

The project was implemented in Python 3.11 with pandas and NumPy for data processing, scikit-learn for scaling, clustering and validation, joblib for model persistence, Plotly for interactive charts and Streamlit for the user interface (Pedregosa et al., 2011). The implementation follows a modular pipeline so that each stage can be run, tested and replaced independently.

Module	Responsibility
src/clean_data.py	Loads raw data; resolves positions; creates eligibility flags and per-90 features; separates outfield players and goalkeepers.
src/feature_engineering.py	Audits features; performs position-median imputation and position-specific standardisation; saves scalers.
src/train_kmeans.py	Evaluates K = 2...8; trains final position models; saves evaluation tables, cluster labels and centroids.
src/validate_clusters.py	Interprets cluster differences; names archetypes; identifies representative players; tests seed stability.
src/player_similarity.py	Computes Euclidean distances, similarity scores, strengths, weaknesses and nearest-player recommendations.
src/game_engine.py	Generates model-backed questions, difficulty-aware distractors

Module	Responsibility
	and relative scoring.
src/player_pools.py	Curates featured target pools for Star Match while retaining the full database for Scout Mode.
app.py	Provides the premium Streamlit interface, three game modes, four difficulty levels and final scout reports.
src/main.py	Runs the complete preprocessing, modelling and validation pipeline in one command.

4.1 Similarity and Game Scoring

For a selected target, the system compares candidates from the same broad position using Euclidean distance over the standardised feature vector. The reusable similarity score is:

Similarity = 1 / (1 + Euclidean distance)

Within each game round, the strongest displayed candidate receives a relative match index of 100%. A near-best choice receives partial credit:

Round match index = (chosen similarity / best displayed similarity) x 100

This relative index is explicitly labelled as a game measure, not a probability. Exact-match, difficulty and streak bonuses are then added to the round score.

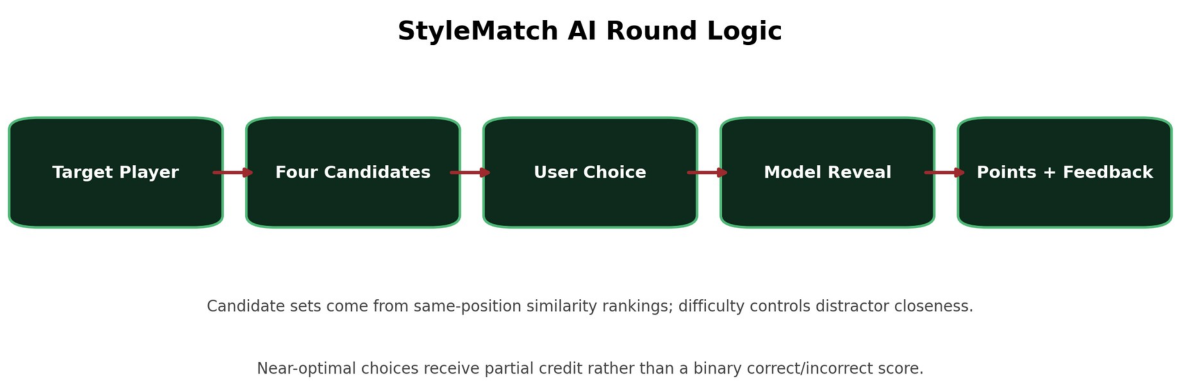

Figure 3. Model-backed logic for one StyleMatch AI round.

4.2 Scope Refinement

The initial proposal also anticipated a manually weighted overall rating and an external LLM explanation layer. To maximise reliability within the course timeline, the final core prioritised a complete clustering-to-interface pipeline. The working game uses transparent relative similarity scoring and deterministic, evidence-grounded explanations; a fully calibrated weighted score and external LLM API remain modular extensions rather than hidden dependencies.

5. Results and Evaluation

5.1 Model-Selection Results

Position	Selected K	Silhouette	Calinski-Harabasz	Davies-Bouldin	Cluster Sizes	Balance Ratio
Defender	2	0.226	113.54	1.858	134 / 395	0.339
Midfielder	2	0.238	236.85	1.619	268 / 436	0.615
Forward	2	0.163	52.50	1.935	94 / 137	0.686

K = 2 produced the highest silhouette score for each position and also compared favourably on the other indices. The scores are modest rather than perfect, indicating that football profiles overlap and that broad position labels do not create sharply separated natural classes. This is an important limitation, but the consistent relative advantage of K = 2, adequate cluster sizes and stable interpretation justify it as a transparent baseline.

5.2 Interpretable Archetypes

Position	Cluster	Archetype	Players
Defender	0	Attacking / High-Involvement Defender	134
Defender	1	Defensive-First Defender	395
Midfielder	0	Attack-Minded Creative Midfielder	268
Midfielder	1	Ball-Winning Midfielder	436
Forward	0	Central Goal-Threat Forward	94
Forward	1	Creative / Wide Forward	137

Centroid analysis gave each group a football interpretation. Attacking defenders showed greater shot, cross, shot-on-target and fouls-drawn activity, while defensive-first defenders recorded lower attacking involvement. Creative midfielders were distinguished by shooting, crossing and attacking involvement, whereas ball-winning midfielders showed higher interceptions and tackles won. Goal-threat forwards had stronger shooting and scoring rates; creative/wide forwards had greater crossing and fouls-drawn activity.

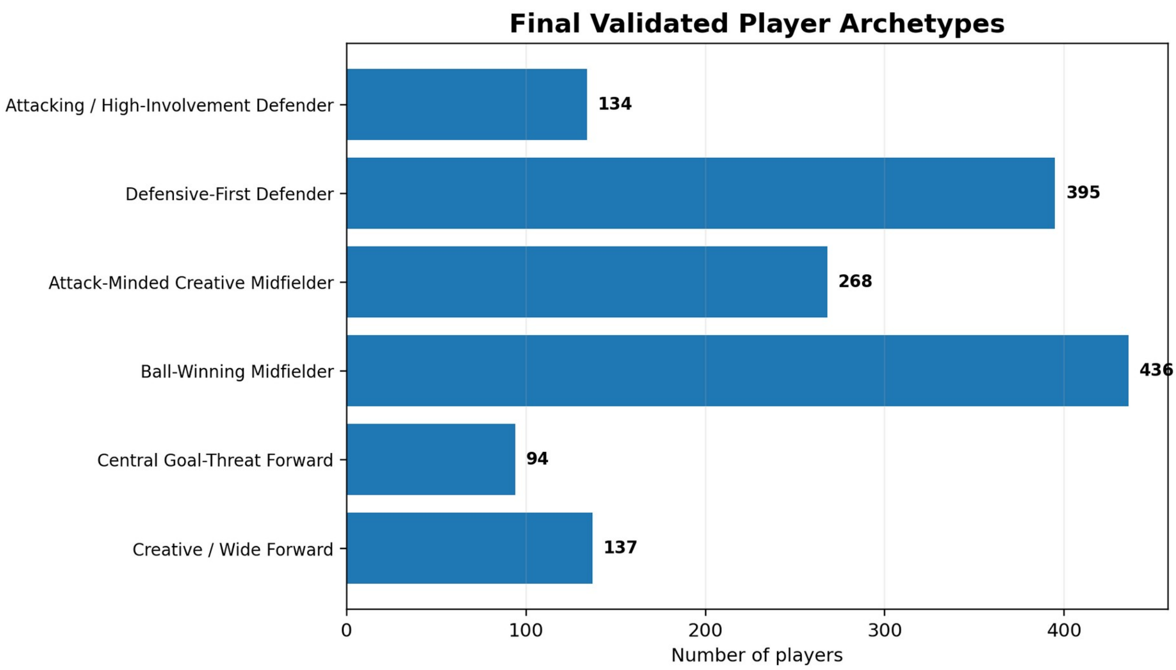

Figure 4. Distribution of the six validated player archetypes.

5.3 Stability, Similarity and Prototype Evaluation

The final position models were rerun with random seeds 0, 1, 7, 21, 42, 77, 100 and 2026. All 24 comparisons achieved ARI = 1.0. This demonstrates that, with the chosen initialisation settings and feature space, the final assignments were reproducible across the tested seeds. It does not prove that the archetypes are the only correct football interpretation.

The similarity engine was tested through direct player searches and successfully returned same-position, same-cluster alternatives together with strengths, weaknesses and distances. StyleMatch AI then operationalised these outputs in a ten-round Streamlit game. Three modes were implemented: Star Match prioritises recognisable targets; Scout Mode uses the full eligible database; Blind Scout hides candidate identities to reduce reputation cues. Rookie, Scout, Analyst and Elite levels alter how close the distractors are to the best match.

Prototype Criterion	Observed Outcome
End-to-end execution	The complete pipeline runs from raw data to validated clusters using python src/main.py.
Question generation	One target and four model-backed candidates are generated dynamically.
Difficulty control	Candidate ranking bands produce progressively closer distractors.
Explanation	The interface reveals the best option, relative choice quality, aligned features and limitations.
Reproducibility	requirements.txt, modular scripts, saved outputs and GitHub branch history support rerunning.
Deployment readiness	The Streamlit application runs locally and is structured for Community Cloud deployment.

6. Ethical Considerations

The ethics audit considered bias, fairness, privacy and transparency. The most significant risk is not direct personal-data exposure but inappropriate confidence: a club could interpret statistical similarity as evidence of equivalent ability, tactical fit or future performance. The interface therefore frames recommendations as decision support and states that live scouting, medical assessment and professional judgement remain necessary.

Risk	Project Response	Residual Limitation
Reputation and identity bias	Names, nationality, club prestige, market value, salary and popularity are excluded from clustering. Blind Scout hides identity during selection.	Performance data still reflect unequal opportunity, team strength and league context.
Positional unfairness	Players are standardised within broad position groups and goalkeepers are separated.	Broad labels cannot distinguish every detailed tactical role.
Small-sample distortion	A 900-minute threshold is used; alternative thresholds are documented.	Injuries and role changes may still make one-season data unrepresentative.
Transparency	Features, formulas, validation metrics, archetype labels and caveats are visible.	Cluster names remain human interpretations rather than objective ground truth.
Privacy	Only public season statistics are used; no sensitive health or personal data are processed.	Dataset licensing and source attribution must still be respected.
Automation bias	The game rewards alignment with the model but explicitly states that the model is not absolute football truth.	Users may still overvalue numerical output if limitations are ignored.

The project also declares the use of generative AI tools. ChatGPT supported brainstorming, code structuring, debugging, documentation and interface design. Generated suggestions were reviewed, executed, corrected against real outputs and edited by the team. The appendix records the purpose of AI assistance and the verification process, in line with the course policy.

7. Limitations and Future Work

- The dataset records broad positions (DF, MF and FW), so the model cannot reliably separate centre-backs, full-backs, wingers and central strikers.
- Per-90 rates do not fully control for team possession, competition strength, tactical instructions, opponent quality or game state.
- The current feature set omits richer event and spatial variables such as progressive passing, pressures, carries and off-ball movement.
- Silhouette scores show meaningful but overlapping clusters; the archetypes should be interpreted as broad profiles, not definitive labels.
- The model is trained on one season from Europe's five major leagues and may not transfer directly to youth, women's, Ghanaian or lower-league football.

- The current prototype does not use injury, medical, personality, contract, salary or transfer-value data.

7.1 Future Work

- Use event-level or tracking data to create detailed tactical roles and contextualised possession-adjusted features.
- Build and validate a separate goalkeeper model.
- Add multi-season and academy data for development tracking and digital player passports.
- Integrate a coach-philosophy module that matches positional player profiles to observed team styles.
- Conduct a formal usability study with scouts, coaches and students, measuring task completion, comprehension and decision quality.
- Calibrate a position-weighted score with football experts or external criteria rather than arbitrary weights.
- Add an optional constrained LLM explanation layer whose output is validated against the supplied statistics.
- Deploy the application publicly and add automated tests and continuous integration.

8. Conclusion

StyleMatch AI demonstrates a complete and responsible application of unsupervised machine learning to football-player analysis. The project moved from raw season statistics to cleaned per-90 features, position-aware standardisation, separate K-Means models, multi-metric validation, interpretable archetypes, a nearest-player similarity engine and an interactive game. The final baseline identifies six broad player profiles and produces stable assignments across the tested random seeds. Its principal contribution is not a claim that clustering can replace football expertise, but a transparent method for converting complex performance data into evidence that users can inspect, challenge and discuss.

The game interface strengthens the system by making human-AI comparison observable: users receive partial credit for near-optimal choices and a final scout report showing how closely their intuition aligned with the model. Within the constraints of broad positional data and a one-season sample, the project satisfies its core objective: it delivers a working, reproducible and ethically framed AI application that supports more structured player comparison while preserving the need for human judgement.

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Appendix A: AI Use Declaration

ChatGPT was used as a development-support tool for system architecture, code templates, debugging, documentation, report structuring and Streamlit interface design. It was not treated as an authoritative source of football facts or model results.

Purposes

- Brainstorming a reproducible data and model architecture.
- Suggesting Python structures for cleaning, clustering, validation and similarity search.
- Supporting debugging of Python environments, Git, Streamlit and deployment.
- Improving documentation, ethical analysis and presentation structure.

Verification and Editing

- All code was run locally rather than accepted as text-only output.
- Syntax and runtime errors were corrected through repeated testing.
- Dataset counts, feature names, K values, cluster sizes and validation scores were checked against generated CSV outputs.
- Unsupported claims were removed; ARI was reported as stability rather than accuracy.
- The final architecture and scope decisions remained the responsibility of the student team.

Appendix B: Team Responsibilities

Team Member	Primary Responsibility from Proposal
Elikem Frank Ameyibor	README documentation and lead presentation.
Joel Elorm Nhyiraba Agyekum	Position-based scoring design and evaluation.
Samuel Asare Opong	Application/front-end development.
Wun-nam Kyirifo Haruna	Player-data collection and cleaning; K-Means clustering and similarity model.

The final integrated prototype extends these roles through shared GitHub collaboration, model-to-interface integration, game design, testing, documentation and presentation preparation.